

## MARCH 2023

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6(b)     | <p>Any <b>two</b> from:</p> <ul style="list-style-type: none"><li>• The design of robots (to perform tasks/operations/functions)</li><li>• The construction of robots (to perform tasks/operations/functions)</li><li>• The operation of robots (to perform tasks/operations/functions)</li></ul>                                                                                                                                                                                                                                                                                                            | <b>2</b> |
| 6(d)(i)  | <p>Any <b>three</b> from:</p> <p>e.g.</p> <ul style="list-style-type: none"><li>• Collects data</li><li>• Stores rules for using the data</li><li>• The ability to reason</li><li>• The ability to learn // uses machine learning</li><li>• ... by adapting what it does</li><li>• ... for example, from mistakes to not make them again // result from previous decisions impacts future</li><li>• ... by changing its own rules</li><li>• ...by changing its own data</li><li>• ...by being trained</li><li>• Makes one or more predictions (to make a decision)</li><li>• Find/analyse patterns</li></ul> | <b>3</b> |
| 6(d)(ii) | <p><b>Four</b> from:</p> <p>e.g.</p> <ul style="list-style-type: none"><li>• Use <b>machine learning</b> algorithms</li><li>• Collects data about where it has been</li><li>• Collect data about obstacles/problems</li><li>• Store successful actions</li><li>• Stores unsuccessful actions</li><li>• Identify/store patterns</li><li>• ... to make sure it does not repeat the same incorrect route</li><li>• ... so, it knows how to react to obstacles next time</li><li>• ...so, it knows what is most likely to work next time</li></ul>                                                               | <b>4</b> |

## JUNE 2023

| Question | Answer                                                                                                                                                                                                                                                                                                               | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 9(a)     | Any <b>two</b> from: <ul style="list-style-type: none"><li>• It has a <b>mechanical</b> structure/framework</li><li>• It has <b>electrical</b> components // by example</li></ul>                                                                                                                                    | 2     |
| 9(b)     | Any <b>two</b> from:<br>e.g. <ul style="list-style-type: none"><li>• Employees don't need to lift heavy furniture</li><li>• Employees can be protected from dangerous tasks</li><li>• Employees can utilise their skills in other tasks</li><li>• Employees don't need to perform repetitive/mundane tasks</li></ul> | 2     |

| Question | Answer                                                                                                                                                                                                                                                          | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 9(c)     | Any <b>one</b> from:<br>e.g. <ul style="list-style-type: none"><li>• Expensive to <b>install/purchase/setup</b></li><li>• High ongoing costs/maintenance costs</li><li>• May deskill the workforce</li><li>• If they malfunction, production may stop</li></ul> | 1     |

## JUNE 2023 (2)

| Question | Answer                                                                                                                                                                                                                          | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 9(a)     | <b>Three</b> from: <ul style="list-style-type: none"><li>• Rule base</li><li>• Knowledge base</li><li>• Interface</li></ul>                                                                                                     | 3     |
| 9(b)     | Any <b>two</b> from: <ul style="list-style-type: none"><li>• It makes decisions</li><li>• ... by <b>applying</b> the <u>rules/logic</u> to the <u>facts/knowledge</u> ...</li><li>• ... to provide a result/diagnosis</li></ul> | 2     |

## JUNE 2023 (3)

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| Question | Answer                                                                                                                                                                                                             | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7(a)     | <ul style="list-style-type: none"><li>• Interface</li><li>• Knowledge base</li></ul>                                                                                                                               | 2     |
| 7(b)     | Any <b>two</b> from: <ul style="list-style-type: none"><li>• <b>Stores</b> the rules for the system</li><li>• ... for the inference engine to use</li><li>• Used to link the facts in the knowledge base</li></ul> | 2     |

## NOVEMBER 2023 (2)

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10(a)    | <p>Any <b>one</b> from:</p> <ul style="list-style-type: none"> <li>– The ability to learn/adapt // machine learning abilities</li> <li>– The collection of data and the rules for using that data</li> <li>– The ability to reason // has problem solving abilities // makes predictions</li> <li>– Simulates intelligent/human behaviour</li> <li>– Analyses patterns</li> </ul>                                                        | <b>1</b> |
| 10(b)    | <p>Any <b>six</b> from:</p> <ul style="list-style-type: none"> <li>– It has an interface ...</li> <li>– ... used to input data/view output</li> <li>– It has a knowledge base</li> <li>– It has a rule base</li> <li>– It has an inference engine</li> <li>– Applies the rule base to/and the knowledge base to provide output/diagnosis/result/solution/decision</li> <li>– Decides what to ask next based on the data input</li> </ul> | <b>6</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11(a)    | <ul style="list-style-type: none"> <li>– Amount of liquid/gas/steam flowing/moving through an environment</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>1</b> |
| 11(b)    | <p><b>Two</b> from (for benefit and matching description)<br/>e.g.:</p> <ul style="list-style-type: none"> <li>– Increases safety</li> <li>– ...meaning that workers do not need to go into dangerous areas to collect data/make checks/do dangerous tasks</li> <li>– Can increase jobs/skills</li> <li>– ...as employees are needed to learn/maintain the equipment</li> <li>– No need to do repetitive tasks</li> <li>– ... so, they can use their time on other/more skilled tasks</li> </ul>                                                                                                                                                                                        | <b>2</b> |
| 11(c)    | <p><b>Two</b> from (for drawback and matching description)<br/>e.g.:</p> <ul style="list-style-type: none"> <li>– High <b>set-up/installation</b> costs</li> <li>– ... it would mean the company need to find a lot of money up front to pay for the equipment // employees will need training</li> <li>– Utility/maintenance/repair costs</li> <li>– ... increase in bills such as electricity // skilled employees will be required to maintain the system // equipment will break/need updating</li> <li>– Deskilling of the workforce</li> <li>– ... may mean that workers will no longer have the skills for some of the manufacturing jobs, should the equipment break</li> </ul> | <b>2</b> |

## NOVEMBER 2023 (3)

| Question | Answer                                                                                                                                                                                                                                                   | Marks    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10(a)    | <b>One</b> mark for each correct term, in the correct order: <ul style="list-style-type: none"><li>– Knowledge base</li><li>– Inference engine</li><li>– Rule base // knowledge base</li><li>– Knowledge base // rule base</li><li>– Interface</li></ul> | <b>5</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10(b)    | Any <b>four</b> from:<br>e.g. <ul style="list-style-type: none"><li>– It is a form of artificial intelligence</li><li>– Means it can adapt/change (its own processes) // It can edit its own algorithms</li><li>– It can edit its own data</li><li>– It can be trained</li><li>– ... this can be supervised/unsupervised</li><li>– ... meaning it can learn with/without human interaction</li><li>– Analyses patterns and stores <b>successful/unsuccessful</b> results ...</li><li>– ... to influence future decisions</li><li>– (Supervised) means a user tells the system the input and output</li><li>– (Unsupervised) means the system is given the input and needs to work out the output</li></ul> | <b>4</b> |